

PROFESSIONAL PROFILE

I combine deep pre-AI systems engineering (5+ years in C/C++, Linux kernel internals, DPDK/VPP packet processing, and hardware) with modern AI tools (OpenAI Codex Desktop, DeepMind AGY CLI). In my projects, AI is never a fragile prompt gimmick; it is an intelligent cognitive brain orchestrated by a robust, deterministic software layer and production bare-metal Linux infrastructure.

CORE COMPETENCIES & TECHNICAL STACK

Systems & Low-Level: C, C++20, Linux Kernel, Device Drivers, MIPS, STM32, DPDK, VPP, DPI, conntrack, POSIX, CMake, Make	Backend & Databases: Python (FastAPI, asyncpg, Pydantic), Node.js (Fastify, Express), PostgreSQL (18+ migrations), Redis, SQLite
Frontend & Mini Apps: React 19, TypeScript, Telegram Mini Apps API, Tailwind CSS, Vite, Framer Motion, SEO, Core Web Vitals	Infrastructure & AI: Bare-Metal Ubuntu, Docker, Docker Compose, Nginx, Caddy, Cloudflare Tunnels, systemd, Codex Desktop, AGY CLI

SELECTED FEATURED PROJECTS

Emily Brautmoden Boutique — Premium bridal boutique redesign & commercial showcase in Germany	React 19, TypeScript, Vite, i18n
React 19, TypeScript, Vite, modular CSS with custom token-driven design system, Framer Motion, react-i18next (DE/EN/RU), react-router-dom v7, Core Web Vitals optimization (LCP, CLS), deployed on Vercel/Netlify.	
Shermos Bot & Mini App — Telegram service & Mini App for manufacturing company in Bishkek	FastAPI, PostgreSQL, Redis, React 19
Webhook + Python worker architecture with FastAPI, asyncpg, Pydantic. Distributed Redis queues with per-user mutex locks. 18 PostgreSQL migrations. 3D rendering pipeline. Telegram Mini App in React 19, TypeScript, and Vite.	
Musikschule CMS Bielefeld — Web platform for music academy in Bielefeld, Germany	React, Vite, Tailwind, Nginx
React + Vite + Tailwind + Framer Motion. Image optimization pipeline with AVIF/WebP and responsive srcSet. Self-hosted Nginx, automated Let's Encrypt (ACME) certificates, HTTP/2, HTTP/3, and edge caching.	
"I Can Do It Myself!" — Children's Workshop — Interactive platform & life-skills workshop program for children	React, Vite, Tailwind CSS, Netlify
Lightweight responsive web application built with React and Vite, optimized for mobile usage by event leads and parents, fast static delivery via Netlify CDN.	
New Testament Theology (Thomas Schreiner) — Bilingual academic reader for Schreiner's work, quotes index & audiobook	React 19, TypeScript, Tailwind CSS, OCR / TTS
React 19 + TypeScript + Tailwind CSS, automated content generation scripts (generate-reader-artifacts), ingestion service for OCR/TTS processing, reference-driven UI tokens, Netlify deployment.	

PROFESSIONAL WORK EXPERIENCE

Software Engineer / Independent Contractor | Freelance & Own Projects

Mar 2025 – Present · Remote (Clients from Germany, Kazakhstan, Kyrgyzstan)

- Infrastructure provisioning: AWS EC2, bare-metal Ubuntu servers, Docker / Docker Compose, Nginx, Caddy, systemd units.
- Networking & Security: automated Let's Encrypt (ACME) certificates, Cloudflare tunnels, HTTP/2, HTTP/3, and API edge protection.
- DevOps workflows: idempotent Bash deployment automation, automated PostgreSQL migrations, monitoring, and production profiles.
- Built production Telegram Mini Apps with FastAPI, asyncpg, Redis distributed locks, and React 19.

Programming Languages Instructor (C / C++) | Luxkod

Oct 2023 – Aug 2024 · Kazakhstan

- Authored and delivered hands-on C and C++ curricula: pointers, memory management, data structures, algorithms.
- Guided students through compiler error diagnosis, memory leak profiling (Valgrind, GDB), and modular project architecture.

Software Engineer / Automation Manager | TOO Zavod Armatura Taraz

Jan 2023 – Oct 2023 · Taraz, Kazakhstan

- Engineered a dedicated C++ application for automated extraction and generation of production reports from Excel datasets.
- Created VBA automation pipelines for inventory reconciliation, logistics manifests, and delivery contracts.
- Streamlined end-to-end document workflows for cargo logistics and warehouse distribution.

C Developer / Systems Engineer | Ideco

Nov 2022 – Dec 2022 · Ekaterinburg

- Developed C systems software leveraging DPDK and VPP (Vector Packet Processing) network stacks.
- Engineered control-plane API extensions and automated unit tests for DPI (Deep Packet Inspection) modules.
- Investigated and resolved bugs in traffic-filtering engines: IPS, DPI, and Linux kernel conntrack.
- Maintained CMake build systems and collaborated in Jira and Git.

Systems & Embedded Software Engineer | In-Soft LLC

Aug 2021 – Oct 2022 · Novosibirsk

- Patched Linux kernel and developed hardware device drivers to expand wireless device capabilities.
- Authored userspace utilities and Bash scripts for dynamic access point configuration.
- Implemented networking services: DHCP, RADIUS, TFTP, SSH, Docker environments, and config parsing via json-c.

Laboratory Assistant, 3rd Category | Budker Institute of Nuclear Physics (BINP SB RAS)

Oct 2020 – Aug 2021 · Novosibirsk

- Programmed STM32 microcontrollers in C for real-time digital noise filtering from electret microphones.
- Implemented digital signal processing algorithms interfacing with sensitive analog sensors.

EDUCATION

Novosibirsk State University (NSU)

2018–2021 | Новосибирск

Physical & Technical Informatics (2018–2021) · Strong mathematical & physics foundation